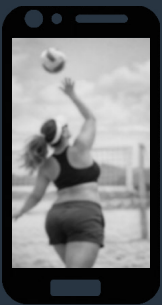
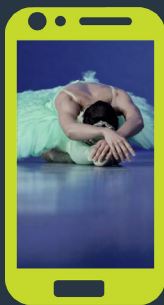
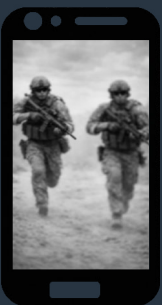
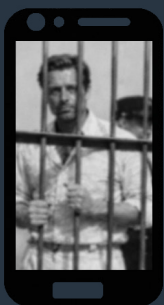
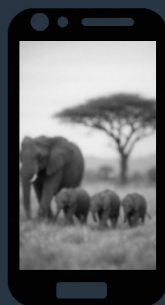
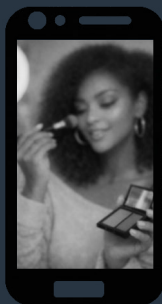
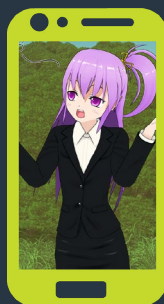
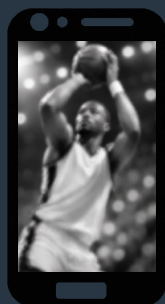


# MOBILE PERSONALIZATION

The history and theory of  
mobile streaming media



JUSTIN GRANDINETTI  
CHARLES ECENBARGER

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*The History and Theory of  
Mobile Streaming Media*

by  
Justin Grandinetti &  
Charles Ecenbarger

# Mobile Personalization: The History and Theory of Mobile Streaming Media

by Justin Grandinetti & Charles Ecenbarger

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## CHAPTER FOUR

# The Characteristics of Mobile Streaming

Streaming audio-visual media is popularly tied to Web 2.0, the era known as the participatory web.<sup>1</sup> The mid-2000s were a time where what we'd now consider relatively rudimentary versions of some still-stalwart websites originated; platforms like Facebook, Twitter, Wikipedia, and YouTube made their debuts alongside less popular, shorter-lived platforms like Foursquare, Tumblr, and Skype. Netflix as a company predates Web 2.0, originating in 1997 as a mail-delivery DVD rental service. It was in 2007 that Netflix launched its now ubiquitous streaming platform, which, along with YouTube and a growing number of competitors, added streaming media to this changing digital landscape. Contemporarily, streaming audio-visual content has become intertwined with countless platforms and websites. If one could go back in time and use the past versions of these platforms, the social networks of the early participatory web would now seem immensely dull in a landscape of the endless scroll, swipe, and stream of content. Audio-visual content is embedded in just about any website one visits—even if it is in the form of irritating auto-play popup ads. In short, if the early 2000s internet is defined by participation and user-generated content, then the contemporary web is all about streaming. One major challenge of the proliferation of streaming is that the term itself begins to lose some conceptual specificity. That is, it was easier to define streaming media when the practice was limited to a few major providers like Netflix,

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<sup>1</sup> Gerard Goggin, "Mobile Web 2.0: New Imaginaries of Mobile Internet," in *Theories of the Mobile Internet: Materialities and Imaginaries*, ed. Andrew Herman, Jan Hadlaw, and Thom Swiss (Routledge, 2015); Tim O'Reilly, "What Is Web 2.0: Design Patterns and Business Models for the Next Generation of Software," *Communications and Strategies*, no. 65 (2007).

Hulu, YouTube, etc. Now that streaming has become entangled and embedded with all types of sites, platforms, and mobile devices, the concept itself becomes opaquer.

We've made the case for a more expansive history of mobile streaming media, what we call mobile personalization, in chapter 1 in tracing some audio-visual antecedent technologies that provide a mobile and portable orientation to entertainment, information, and monetization. In a narrower and more linear sense, streaming is often connected to the first livestreamed radio broadcast by ESPN SportsZone of a baseball game on September 5, 1995.<sup>2</sup> This broadcast was audio-only, yet this event marked the beginning of a series of legal battles for ownership of streaming media technologies. Streaming functions much like loading a website, in that the process does not involve a permanent download. We engage online through what's known as packet switching, in which a message is broken into smaller parts for transfer then reassembled at the destination.<sup>3</sup> Packet switching works on the distributed network of the internet because packets can take the quickest of multiple routes to their destination, which allows the network increased resilience. TCP/IP protocols establish the rules of connection on the web, creating a form of standardization in the way packets are transmitted, routed, and received.<sup>4</sup> While packet switching functions because packets can arrive in any order, streaming media requires a temporal continuity. Streaming video, then, utilizes additional protocols to ensure packets arrive and are assembled in the correct order, including real-time transfer protocol (RTP), real-time streaming protocol (RTSP), and real-time transport control protocol (RTCP), and HTTP-based adaptive streaming.<sup>5</sup> Collectively, one way to define streaming is strictly technological, in that audio-visual streaming is based off layered transfer technologies and protocols.

Of course, we've worked to establish throughout this book that streaming is not just technical but sociotechnical; it is more than merely platforms, infrastructures, code, and devices. Our mobile personalization approach highlights how streaming is a major shift in how we watch, what we watch, and how we relate and connect to one another. It's also tied to major indus-

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<sup>2</sup> Alex Zambelli, "A History of Media Streaming and the Future of Connected TV," *Guardian*, March 1, 2013.

<sup>3</sup> Katie Hafner and Matthew Lyon, *Where Wizards Stay up Late: The Origins of the Internet* (Simon & Schuster, 1996); Janet Abbate, *Inventing the Internet* (MIT Press, 1999), 34.

<sup>4</sup> Alexander R. Galloway, *Protocol: How Control Exists After Decentralization* (MIT Press, 2004), 8.

<sup>5</sup> Chris Pollette, "How Streaming Video and Audio Work," *How Stuff Works*, May 1, 2024.

try shifts, wherein a mix of long-tail niche markets, digital piracy, changes in ownership by artists and creators, increased power of distributors, and data-driven infrastructures came together to disrupt traditional content distribution models for music, film, and television alike.<sup>6</sup> In essence, defining streaming is a challenge. As explained by Ghislain Thibault, “For the television industry and television audiences alike, the term ‘streaming’ seems to mark the transition from analogue to digital television and the new types of small-screen consumption that accompany it.”<sup>7</sup> Other scholarship has attempted to address what exactly is novel (or remediated) about this distribution model. Some focus on changing notions of live content, or the altered shared experience formerly felt in mass media. Others emphasize the dimension of time, in how streaming permits what’s known as time-shifting to alter rhythms of traditional viewing to fit the user. When it comes to new forms of mobile streaming, scholarship on hybridity of the physical, digital, and malleable spatial and temporal nature of screens is more relevant than ever. And it’s impossible to leave out the elements of data collection and algorithmic recommendation—for fixed or mobile streaming.

Existing scholarly accounts that attempt to make sense of the socio-technical shifts of streaming media fall back on how the medium alters or functions as a continuation of traditional audio-visual media habits. And that makes sense. Streaming in the era of Web 2.0 often took place in the fixed site of the living room on a web-connected television. Accordingly, streaming initially functioned as a direct parallel with the televisual. But this these practices have rapidly changed. Scholarship about streaming as spatially fixed has provided excellent conceptual anchors, yet it is critical to reassess and reconceptualize these attributes in the era of mobile streaming. In this final chapter, we set out to define the key characteristics of mobile streaming. In doing so, we engage with the question of how mobile streaming remediates prior understandings of flow, liveness and sharedness, screens and space, mobile and hybrid, and algorithms and big data. These concepts are all critical to understanding how mobile streaming is part of a longstanding trajectory of social, technical, and infrastructural shifts.

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<sup>6</sup> Michael D. Smith and Rahul Telang, *Streaming, Sharing, Stealing: Big Data and the Future of Entertainment* (MIT Press, 2016), 56.

<sup>7</sup> Ghislain Thibault, “Streaming: A Media Hydrography of Televisual Flows,” *View* 4, no. 7 (2015): 110.

## *Mobile Streaming Flow*

In 2016, the social media platform Instagram introduced “Instagram Stories,” a feature that allows users to share photos and videos in slideshow format. In contrast to the other content posted to Instagram, stories create a sequence that users can add to over the course of a day by using text and drawing tools to create overlap. As the user selects the story icon in the Instagram app, stories created by individuals, celebrities, organizations, and brands the user follows on Instagram are shown in unbroken succession, occasionally interspersed with advertisements based on Instagram’s location-aware model of data collection. As these stories are updated throughout the day, users are shown only new content that is added (and of course, new advertisements).

There is an inherent ephemerality to Instagram stories, as this content “disappears”<sup>8</sup> after twenty-four hours, no longer accessible to other platform users. Instagram’s story feature is not necessarily a wholly original concept: Social media platforms such as Vine (a short-form video platform popular from 2012–2016) and Snapchat (a social media platform created in 2011 that allows users to share story-type content that similarly vanishes after a single viewing) provided the model for Instagram to appropriate in its own platform. At the same, what is relevant about Instagram’s co-option of the story feature is the platform’s reach and popularity. Instagram is among the top five most popular social media platforms in the world in terms of monthly users.<sup>9</sup> The success of Instagram stories seems to coincide with users leaving Snapchat.<sup>10</sup> Facebook’s billion-dollar acquisition of Instagram in 2012 is part of this platform recentralization as well, in that users can cross-share their stories to both social media sites.

Instagram stories are not “streaming media” in the colloquial sense, in that they are not the same as the streaming audio-visual media often positioned as part of a film and televisual lineage. Yet, simultaneously, these stories create a streaming “flow” that is influenced by antecedent communication technologies. Now, a few years later, Instagram has launched “Reels,” a direct short-form streaming competitor to TikTok. In a matter of fourteen years, Instagram morphed from a photo image-based platform to a streaming

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<sup>8</sup> It is well established that social media platforms collect and store even temporary and deleted content. Many, like Facebook and Instagram, store this content permanently.

<sup>9</sup> “Most Popular Social Networks Worldwide as of February 2025, by Number of Monthly Active Users,” *Statista*, February 2025.

<sup>10</sup> Sara Salinas, “Peak Social? The Major Social Platforms are Showing a Significant Slow-down in Users,” *CNBC*, August 8, 2018.

one. In essence, the platform, like many others, has attempted to recapture and reimagine flow.

Here, the work of Raymond Williams yet again functions as a conceptual mooring in considering the lineage of our contemporary streaming landscape. Williams explains that screen size, interruptions, participation, and a sequence of rhythms were all aspects of prior media forms remediated by television.<sup>11</sup> Pre-televisual habits and vocabulary, nevertheless, were shaped by the experience of discrete, isolated, and temporary events.<sup>12</sup> It is the concept of flow—how the sequence of program content and commercials create the real and unbroken experience of watching television as unbroken broadcast—that marked a critical departure for this cultural technology. The experience of television, then, captures viewers through this flow and readies them for the planned interruptions of commercials. Flow is a question of power via altered temporal rhythms and a capture of attention. This concept is something that is very much alive in mobile streaming as well.

That being said, flow is an inherently slippery concept—something that's perhaps more sensed and felt than measured. Flow explains the difference between watching a play and the novel experience of the televisual; for example, while a play has a planned intermission that functions to break up content, the televisual instead operates through an industrialized organizational schema in which planned interruptions create an unbroken experience of watching content. We could perhaps quantify flow by studying the temporal patterns of content and advertising over time. What's harder to capture, however, is the massive experiential change that accompanied the arrival of televisual broadcasting and the decades that followed. And the novel forms of mobile streaming add in aspects that were completely foreign to the early televisual era; location-awareness, algorithms, direct user participation and curation, temporal duration, and mobility remediate the familiar experience of flow. Perhaps one of the reasons that flow as a concept endures despite these changes is that Williams himself conceived of flow as flexible and mutable, in that he believed television itself would be subject to technological convergence.

Scholars have taken heed and updated Williams's enduring concept. Williams conceived of flow as a more passive process for the viewer, whereas streaming often requires a push-pull between suggested content and a more

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<sup>11</sup> Raymond Williams, *Television: Technology and Cultural Form* (Fontana, 1974), 87–88.

<sup>12</sup> Williams, *Television, Technology and Cultural Form*, 87–88.

active selection.<sup>13</sup> Stalwart televisual-style streaming platforms like Netflix function by promoting seemingly limitless choice for audiences, while at the same time narrowing these options via algorithmic and data-driven personalization. Netflix has experimented with automatically playing content (either when users hover over a choice or when they finish episodes). Binge-watching, the act of watching multiple episodes in a row of unbroken content, was a defining quality of the early turn toward audio-visual streaming media. This push-pull between active and passive viewing also applies to data collection, including in the era of the “second screen.”<sup>14</sup> Data collection by streaming platforms entails both active and passive data—what we watch, for how long, where we watch, what device we use, when we pause or give up on content, what we like, what we reshare—the list goes on. In essence, the passivity of Williams’s televisual flow is mediated and remediated by streaming and then more mobile orientations to the process.

While not specific to mobile streaming, Sheenagh Pietrobruno reexamines Williams’s flow in the context of YouTube.<sup>15</sup> In an era of algorithmically suggested content, flow takes on a different meaning. That is, instead of a programmed flow driven by one-way dissemination of content that is directly controlled by broadcasters, YouTube and other streaming platforms do not determine exactly what content is viewed by users. This process instead functions as more of a “staccato flow,” wherein the user self-selects short video content that is shaped by underlying algorithms. YouTube’s algorithms—designed for monetization—can both reinforce dominant cultural narratives and enable alternative perspectives to circulate. Elements like audience engagement, video rankings, and metadata contribute to the visibility of varied cultural representations, indicating YouTube’s potential to both homogenize and expand the kind of content that circulates. Stated differently, flow is altered by streaming in multiple ways; the flow of content looks different, it feels different, and the patterns of “unbroken” content have changed as well. Perhaps most importantly, flow is no longer merely the purview of television producers, but instead algorithmically defined.

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<sup>13</sup> William Uricchio, “The Future of a Medium Once Known as Television,” in *The YouTube Reader*, ed. Pelle Snickars and Patrick Vonderau (National Library of Sweden, 2009), 32.

<sup>14</sup> Hye Jin Lee and Mark Andrejevic, “Second-Screen Theory: From Democratic Surround to the Digital Enclosure,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 40.

<sup>15</sup> Sheenagh Pietrobruno, “YouTube Flow and the Transmission of Heritage: The Interplay of Users, Content, and Algorithms,” *Convergence* 24, no. 6 (2018).

We've highlighted the importance of algorithms and big data to streaming in previous chapters of this book, and we'll come back to the significance of these concepts to mobile streaming in a few pages. Suffice it to say, other scholars have connected streaming media, algorithms, and flow. One such theorization comes from Christopher Cox, who defines the contemporary logics of streaming flow—the televisable and invisual.<sup>16</sup> Televisable emphasizes the importance of search bars, menus, autoplay screens, and other platform elements that foster a sense of user control of streaming while downplaying the latent industrial control by platforms. The invisual refers to unseen functions like code, software, algorithms, and data that pull users toward what appears on their screen. Together, these concepts help complicate aforementioned issues of agency within streaming flow, indicating the complex weave of users, platforms, content, and technology that creates the experience of watching.

Flow is clearly a durable concept of the televisual that is remediated by streaming; but where does the mobile element fit in? Kathleen Oswald and Jeremy Packer update the concept of flow for a mobile, screen-saturated landscape, emphasizing how streaming complicates both televisual and filmic traditions, as well as recent studies of digital media. These scholars instead focus on the emergence of “tele-practices” enabled by shifting formats, devices, and mobilities, where content moves fluidly across multiple screens. In a web-connected world, flow no longer binds viewers to a single broadcast but instead captures desire through advertising and spatial-temporal consumption patterns.<sup>17</sup> Screens structure movement and behavior, making individuals responsible for managing flow. Engaging with Williams's theorization of televisual flow—how disparate content forms a seamless narrative—the authors argue that contemporary flow operates across platforms, keeping subjects “fixed in transmission” rather than fixed on one transmission.<sup>18</sup> What matters is less what we watch than the fact that mobile devices transform reality into a “magical space of mobile desire,” promising empowerment but also demanding self-regulation. Mobile screens are part of a reconfiguration of time and space, but these devices and their attendant practices create a flow in the form of governance.

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<sup>16</sup> Christopher M. Cox, “Programming—Flow in the Convergence of Digital Media Platforms and Television,” *Critical Studies in Television* 13, no. 4 (2018): 439.

<sup>17</sup> Kathleen F. Oswald and Jeremy Packer, “Flow and Mobile Media: Broadcast Fixity to Digital Fluidity,” in *Communication Matters: Materialist Approaches to Media, Mobility, and Networks*, ed. Jeremy Packer and Stephen B. Crofts (Wiley (Routledge), 2010). 278.

<sup>18</sup> Oswald and Packer, “Flow and Mobile Media,” 277.

The wealth of scholarship updating flow largely focuses on fixed streaming, whereas Oswald and Packer instead emphasize the importance of the materiality of screens. Though an interesting and generative account, missing from remediations of flow is the space in between—the practices of mobile streaming. Here again, mobile streaming is not simply technical. These practices certainly are undergirded by devices, infrastructures, and platforms. Yet, mobile streaming is also the content, the connectivity, the interaction, and the sociability of it all. The flow of the early three-channel televisual era is certainly not an experience we are able to go back to and reexperience now in an age of constant connection and information overload. Yet the affective dimension of capturing and keeping audience attention through an unbroken stream is, perhaps, more important than ever.

Flow is not scaffolded by screens; it is accessed in a variety of spaces and fills all sorts of time in between activities. We reach for our phones instinctively. And younger people are most likely to open platforms and begin watching flows of short video content. Flow is also algorithmic and data-driven, another remediated aspect of mobile streaming. In short, contemporary examples—from Instagram stories to TikTok, from popup video ads to embedded video on news sites, and from YouTube Shorts to streaming music platforms—all function as a remediated flow.

To summarize, flow—how the rhythms of content and planned interruptions create an unbroken organizational schema of watching—remains relevant to streaming and mobile streaming. Considering the multifarious, embedded, and ever-changing ways that streaming is now brought to us via our mobile web-connected devices, nevertheless, necessitates a reconsideration of flow. As previously mentioned, flow is something that is experienced. It's hard for someone born in the early 2000s to comprehend flow in the halcyon era of cable television, just as 90s kids had to trust our parents explaining what the early era of the televisual felt like. What is clear is that something is *different* now. Instead of a neatly planned corporate scheme, flow is now structured by algorithms. Instead of existing solely in the domestic site of the home, flow now fits into the cracks of our lives when we mechanically reach for our phones in moments of boredom. And instead of tuning in at set times, we now experience flow in an individual sense. Mobile flow is, accordingly, an essential element present in the televisual through the contemporary mobile landscape, even if it does appear and feel a bit different.

Seemingly at odds with this last point, nonetheless, is the communal element that has been reignited by the personalized and individualized nature of

mobile streaming. Elements of liveness and sharedness—long critical to the televisual—are remediated by streaming and then again by mobile streaming.

### *Liveness, Sharedness, and Mobile Streaming*

The first academic work one of this book's authors ever published is a chapter about the changing experience of communal television in the nascent years of Netflix's streaming platform. Originally conceptualized in 2015, the chapter centers on the relationship that was at that time developing between streaming platforms, audiences, and shared online conversations about content. At the heart of this change was Netflix upending the traditional content release model by providing viewers "all at once" access to entire new seasons of the platform's original programming like *Arrested Development*, *House of Cards*, and *Orange Is the New Black*. This release model turbocharged the idea of binge-watching, with viewers rushing to finish new seasons faster than ever. Audiences seemed to both love and hate this schedule. On the one hand, there was no waiting a week in between new episodes; on the other hand, something seemed "lost" in the lack of anticipation. Websites that covered popular culture often lamented this model too. If one watched half the season the day it was released, were they "ahead" or "behind"? After all, is watching content all that fun if you can't talk about it with others?

Now over a decade later, the traditional televisual release model of weekly new episodes has largely returned. There is likely no single reason for this reversal. For platforms like Netflix, the positive buzz of all-at-once seasons was outweighed over time by the need to keep audiences paying for the platform month after month. Younger readers might not remember the brief period when cord-cutting cable could be achieved by a meager \$10 subscription. The contemporary proliferation of televisual and cinematic subscription platforms now feels like cable costs all over again, and legacy cable providers like Comcast, Spectrum, and Verizon offer their own streaming subscription platforms, joined by cable-like streaming platforms that include Fubo, Sling, YouTube TV, and Hulu Live. Releasing an anticipated new show in a single day means that platform audiences might sign up for a month and then cancel their subscription. From the platform provider perspective, it's better to string them along over ten weeks.

Just as importantly, it's worth considering if binge-watching new content is what audiences truly want. Much has been written about the death of monoculture, or the end of the golden and platinum eras of television.

But it's clear that even now, there is something to the shared experience of televisual-like content. Live sporting events are now viewed via the technical process of streaming, but the togetherness they provide never went away, even as formats changed. The slowed pace of televisual releases similarly works to give audiences a chance to engage in conversations, fan theories, and speculation each week. These conversations, once the purview of comment sections on pop culture think-piece and recap articles, have now migrated to short-form social media streaming videos and podcasts. Streaming about streaming. And that's before considering just how much viewing time of televisual and cinematic content has been outright replaced by streaming on social media platforms, accessed on mobile devices.

In essence, this situation speaks to concepts that are mediated and re-mediated by streaming and then mobile streaming: liveness and sharedness. Put differently, liveness and sharedness are at the elusive heart of streaming. Personalization is desirable for viewers, but too much personalization can be an isolating experience. Clearly, there is something to watching live (or some approximation thereof) and engaging with shared content. And this desire for live and shared connection is remediated by mobile streaming.

Historically, one-to-many communication models facilitated both asynchronous and synchronous interaction. Asynchronous media such as books, newspapers, and early websites allow users to engage with content at their own pace. In contrast, synchronous media such as broadcast radio, television, and cinema demand simultaneous engagement, underscoring the significance of temporal coordination.<sup>19</sup> Within these synchronous formats, concepts such as liveness and sharedness emerge as fundamental to understanding media viewing and engagement patterns. Liveness, as conceptualized by Nick Couldry, plays a pivotal role in reinforcing social coordination around media “centers” that disseminate images, narratives, and information in near real-time.<sup>20</sup> While there are rival forms of liveness, especially when accounting for streaming media, Couldry argues that liveness—in the general sense of continuous connectedness—is likely to remain an essential aspect of new media. This is because media's role in shaping temporal and spatial organization of the world, along with notions of shared reality and what is important, is still valued by users.<sup>21</sup>

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<sup>19</sup> Klaus Bruhn Jensen and Rasmus Helles, “Speaking into the System: Social Media and Many-to-One Communication,” *European Journal of Communication* 32, no. 1 (2017): 17.

<sup>20</sup> Nick Couldry, “Liveness, ‘Reality,’ and the Mediated Habitus from Television to the Mobile Phone,” *The Communication Review* 7, no. 4 (2004): 353.

<sup>21</sup> Couldry, “Liveness,” 357.

Film is an early visual (and then audio-visual) medium that allowed for one-to-many dissemination and shared experience via storage. That is, film is captured and stored for presentation at certain times in limited spaces of the theater. Liveness, in a televisual sense, is connected to notions of simultaneity and storage. But viewed retrospectively, there exist debates on exactly how “live” the televisual ever really was. For instance, William Uricchio contends that the televisual is dominated by a center to periphery model that focuses on either simultaneity (and liveness) or storage.<sup>22</sup> While Uricchio believes simultaneity and storage become conflated via television, Doron Galili instead highlights that the desire for simultaneity and storage oscillates within the development of the televisual, as both are present and separate in a goal of “mastering time.”<sup>23</sup> Regardless of these details, what is clear is that liveness and sharedness have long been a critical aspect of audio-visual media. And elements of liveness and sharedness are deeply entangled with storage, space, and time.

The example of Netflix’s release model highlights a brief period of disrupted liveness and sharedness. As mentioned, more traditional televisual release patterns have seemingly returned, but not before some interesting experiments that highlight how streaming media converged with social media. One experimental feature that brought together the elements of location-awareness, liveness, sharedness, and plasticity is Facebook’s 2018 feature called Watch Party. Watch Party allowed Facebook group administrators to select a public video and share it to other group members to create a simultaneously streaming “shared viewing experience.”<sup>24</sup> According to Facebook Head of Video Fidi Simo, “Watching, commenting and reacting to the same moments together . . . helps build the kind of community and engagement we’ve seen with Live.”<sup>25</sup> Within Watch Party, viewer comments and emoji reactions are overlaid on the video stream, like the setup of mobile Facebook livestreams. This shift aligns with remarks from Facebook’s VP of News Feed, Adam Mosseri, who noted that publisher content distribution would likely decline as the platform reorients itself toward direct social engagement.

Over time, Facebook’s parent company Meta significantly increased its emphasis on live streaming as a core feature, largely in response to com-

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<sup>22</sup> Uricchio, “Television’s First Seventy-Five Years,” 287.

<sup>23</sup> Doron Galili, *Seeing by Electricity: The Emergence of Television and the Modern Mediascape, 1878–1939* (Duke University Press, 2020), 224.

<sup>24</sup> Josh Constine, “Facebook ‘Watch Party’ Lets Groups View Videos Simultaneously,” *TechCrunch*, January 17, 2018.

<sup>25</sup> Constine, “Facebook ‘Watch Party.’”

petitors. Although Facebook is not an exclusively mobile platform, trends indicate that mobile access constitutes its primary mode of engagement. This shift is exemplified by Facebook's 2012 acquisition of Instagram and the subsequent integration of Instagram Stories across both platforms, reflecting the company's strategic investment in mobile streaming technologies. One early and related example of this emphasis comes from Facebook's continued support of Watch Party, wherein the company began testing a simultaneous live viewing feature embedded within the Facebook Messenger just ten months after the rollout of Watch Party.<sup>26</sup> Yet again, this form of streaming engagement is not inherently mobile, but the fact that Facebook overhauled its mobile-oriented Messenger application to include Watch Party shows the increasingly mobile dimension to streaming media. Furthermore, such features emphasize not a death of liveness and sharedness, but new forms of mobile capture of the desire for collective temporal engagement. The Watch Party feature is one strategy to address concerns that the live experience of viewing would be lost via the advent of streaming.<sup>27</sup> Here, sharedness, liveness, mobility, and temporal flexibility meet via new streaming practices.

Research into streaming practices finds that despite the scheduling freedom and time-shifting potential that streaming media provides, viewers remain aware of the viewing habits and patterns of others.<sup>28</sup> Others similarly establish that streaming viewers still desire a sense of togetherness and develop new strategies for social cohesion despite disrupted viewing practices.<sup>29</sup> This so-called plasticity refers to the way that individuals accessing streaming media are able to fill the cracks of their leisure time through the malleable nature of streaming, illustrating how streaming media integrates seamlessly into the fragmented moments of daily life. Examples from the past decade validate these findings in action: The initial disruptions of streaming as something that lacked structured temporality and community have rubber banded back toward something that is both the same and different, particularly when factoring in new forms of mobile streaming. That is to say that while contemporary liveness and sharedness might not look like the nuclear family sitting together in the living room watching three-network television

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<sup>26</sup> Mallory Locklear, "Facebook is Testing a Watch Party-Like Feature for Messenger," *Engadget*, November 16, 2018.

<sup>27</sup> Couldry, "Liveness"; Lilly Irani et al., "Rhythms and Plasticity: Television Temporality at Home," *Personal and Ubiquitous Computing* 14, no. 7 (2010).

<sup>28</sup> Irani et al., "Rhythms and Plasticity," 624.

<sup>29</sup> Yu-Kei Tse, "Television's Changing Role in Social Togetherness in the Personalized Online Consumption of Foreign TV," *New Media & Society* 18, no. 8 (2016): 1550.

options alongside millions of other households, we still cling to elements of shared experiences via what we stream. Well-articulated by Grame Turner, liveness and sharedness still exist “outside the box” of traditional television: “If liveness includes a sense of the shrinking temporal gap between oneself and the rest of the world, as well as a palpable sense of immediacy, then this is something we can find as readily online as in television.”<sup>30</sup> Accordingly, the contemporary landscape of always available mobile streaming connectivity only furthers this notion of liveness.

Streaming and mobile streaming practices are possible through arrangements of technologies, platforms, infrastructures, and individuals that allow temporal and spatial coordination via new forms of liveness and sharedness. Despite initial prognostications that streaming would lead to the death of liveness, along with other consequences for a shared sense of content, mobile streaming reveals how these concepts are more relevant than ever. As streaming has become entangled with social media, it has simultaneously become a means to view and listen to content on the go. And the elements of live viewing and shared experiences have come to form the bedrock of what it means to stream. TikTok and Instagram Reels, among other platforms, are exemplars of a new orientation to streaming that reactivates the televisual and cinematic theories of liveness and sharedness. In the next section, we discuss how screen and spatial scholarship is remediated by mobile streaming.

### *Screens, Space, Streaming*

In past comments about Netflix, both CEO Reed Hastings and Chief Content Officer Ted Sarandos have commented on mobile viewing and screen size, explaining that the company has no plans to hunt for “shorter content or asking producers to change their shows to suit specific viewing situations.”<sup>31</sup> Explicitly stating “We’re not trying to adapt to the size of a screen,” both figures acknowledged that young people tend to watch via mobile devices.<sup>32</sup> However, Hastings was quick to note that users watch content across multiple screens, as the convenience of being able to pause and pick back up with content in another location via a different device is important. Just a few years later, comments in Netflix’s 2019 fourth-quarter earnings report

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<sup>30</sup> Grame Turner, “‘Liveness’ and ‘Sharedness’ Outside the Box,” *Flow* (2011).

<sup>31</sup> Chris O’Brien, “Netflix’s Reed Hastings and Ted Sarandos Explain Why Screen Size Doesn’t Impact Their Shows,” *Venture Beat*, May 4, 2016.

<sup>32</sup> O’Brien, “Netflix’s Reed Hastings.”

stated that “We earn consumer screen time, both mobile and television, away from a very broad set of competitors. We compete with (and lose to) ‘Fortnite’ more than HBO.”<sup>33</sup> Fortnite is a massively popular online multiplayer survival game, available on a variety of devices that includes smartphones. The comment from Netflix articulating Fortnite as a direct competitor shows a breakdown between fixed and mobile, wherein the screen now occupies a variety of spaces beyond the living room. Better understanding this contemporary moment of mobile streaming necessitates a focus on two interrelated aspects of scholarship: space and screens.

Prior to what’s known as the spatial turn of the 1970s and 80s, the concept of space was subordinated to a focus on time; space was often treated as dead, fixed, and immobile, whereas time was rich, lively, and dialectic.<sup>34</sup> Spaces were often conceived of geometrically as an empty area and as a container that was given meaning by the activities taking place there.<sup>35</sup> Scholars like Michel Foucault, Henri Lefebvre, and Edward Soja, among others, instead began rethinking space not as static, but instead lively. Space is a dynamic arrangement, something at once geographic, economic, demographic, sociological, ecological, political, commercial, national, continental, global.<sup>36</sup> Doreen Massey defines spaces as the product of interrelations, as spheres of possibility and heterogeneous multiplicity, and as always under construction.<sup>37</sup> Space, then, cannot be considered static, but instead as an ongoing assemblage of diverse local and global flows.

Critical to these theorists is the notion that space is relational, and that these relations contribute to the construction of values, meanings, knowledge, control, and production. This book emphasizes how the material and discursive work in tandem with one another. One aspect that’s important to note about the spatial turn is that, under these conceptualizations, space is not subject to essentialisms nor ontological dualisms. That is, the materiality of space may be given, but its organization and meaning is a product of dynamic and ongoing relations.<sup>38</sup> Here, the materiality of space is not

<sup>33</sup> Kevin Webb, “Netflix Says It’s More Worried About Competition From Video Games Like ‘Fortnite’ than Other Streaming Services,” *Business Insider*, January 18, 2019.

<sup>34</sup> Michel Foucault, “Questions on Geography,” in *Power/Knowledge: Selected Interviews and Other Writings, 1972–1977*, ed. Colin Gordon (Pantheon, 1980), 61.

<sup>35</sup> Henri Lefebvre, *Everyday Life in the Modern World* (Routledge, 1984).

<sup>36</sup> Lefebvre, *Everyday Life*, 4.

<sup>37</sup> Doreen Massey, “For Space,” *Geografiska Annaler: Series B, Human Geography* 89, no. 1 (2007): 9.

<sup>38</sup> Edward Soja, *Postmodern Geographies: The Reassertion of Space in Critical Social Theory* (Verso, 1989), 122.

denied, nor is it considered separate from social space; instead, social relations via discursive formations (and technologies) give meaning, definition, and organization to the matter of space.

Insights on how technologies are part of the ongoing organization of space, and by extension, regimes of power and control have been taken up in a wealth of scholarly literature on the cinematic<sup>39</sup> and the televisual.<sup>40</sup> One such specific example comes in James Hay's historical account of the televisual as part of spatial arrangements for the production, maintenance, and management of social relations.<sup>41</sup> Hay considers how, over time, the television is brought into productive relationship with other environmental elements, in turn decentering the television itself while still considering how media matters at various sites. What is key for Hay is how television is both a site and a network, part of sociospatial arrangements of interconnectivity that help manage how individuals sense, experience, and move through the world. In an account more specific to the domestic site of the home, Lynn Spigel notes the way that televisual technology provided new connection and collapsed together what had formerly been separate public and private spaces. With television, the home is no longer divorced from public world; instead, the outside world is filtered in via broadcast technology, albeit in the form of recreation and consumerism.<sup>42</sup> In turn, the relationship between television and space is a question of how regimes of power and control develop and are managed across sites throughout history.

Other more specific accounts on the relationship between the television and public space include Anna McCarthy's examination of television's pervasive role in public space, emphasizing its materiality and its influence on spatial experiences, public-private boundaries, and institutional power structures.<sup>43</sup>

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<sup>39</sup> Bill Marshall, "Montreal Between Strangeness, Home, and Flow"; Adrian Fielder, "Poaching on Public Space: Urban Autonomous Zones"; J. Paul Narkunas, "Streetwalking in the Cinema of the City: Capital Flows Through Saigon"; Rolando B. Tolentino, "Cityscape: The Capital Infrastructuring and Technologization"; and Geoffrey Nowell-Smith, "Cities: Real and Imagined," in *Cinema and the City: Film and Urban Societies in a Global Context*, ed. Mark Shiel and Tony Fitzmaurice (Blackwell, 2001).

<sup>40</sup> Jeremy and Kathleen F. Oswald, "From Windscreen to Widescreen: Screening Technologies and Mobile Communication," *The Communication Review* 13, no. 4 (2010); Paul Vonderau, "Beyond Piracy: Understanding Digital Markets," in *Connected Viewing: Selling, Streaming and Sharing Media in a Digital Era*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014).

<sup>41</sup> James Hay, "Locating the Televisual," *Television & New Media* 2, no. 3 (2001): 212.

<sup>42</sup> Lynn Spigel, *Make Room for TV: Television and the Family Ideal in Postwar America* (University of Chicago Press, 1992), 18.

<sup>43</sup> Anna McCarthy, *Ambient Television: Visual Culture and Public Space* (Duke University Press, 2001), 1.

She argues that television screens are embedded in local power dynamics and everyday environments, shaping public spaces in distinct ways. In later work, McCarthy refines this perspective, conceptualizing the screen not as a static object or network but as something that “takes place” within social and spatial contexts.<sup>44</sup> By contrast, Ginette Verstraete frames screens as active components within heterogeneous power relationships that shape public space through consumer integration and technological assemblages.<sup>45</sup> This distinction between screens as seamlessly embedded in daily life (McCarthy) versus screens as dynamic elements within broader sociotechnical systems (Verstraete) defines a key debate in spatial media studies. The rise of mobile streaming further complicates this discourse, as streaming platforms add another layer of spatial experiences through user practices, infrastructures, and augmented space.<sup>46</sup> These spaces, overlaid with dynamic data flows, intensify surveillance, immersion, and user movement. Mobile streaming, by extension, reinforces a bidirectional but asymmetrical relationship, where users gain new forms of engagement in exchange for their data.

One element that’s crucial to consider is that a mobile device does not automatically equate to streaming “on the go” nor in a variety of spaces. In fact, some of the earliest corporate imaginings of mobile streaming–adjacent practices were, paradoxically, largely confined to the home. Some initial attempts to incorporate the smartphone into audio-visual entertainment came via experiments such as the 2012 iPhone app “Avengers: A Second Screen Experience,” which allowed Marvel fans to sync their Blu-Ray views of *The Avengers* (2012) with additional content via behind-the-scenes videos, games, galleries, and information about the movie accessed simultaneously via their phones. Often dubbed “the second-screen experience,” novel attempts to capture audience attention by combining smartphones and more traditional televisual setups included companion viewing apps, social networking apps, franchise apps, and TV everywhere apps.<sup>47</sup> Second-screen adaptations were often a corporate vision of interactivity that forces audience’s attention and

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<sup>44</sup> Anna McCarthy, “From Screen to Site: Television’s Material Culture, and Its Place, in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 93.

<sup>45</sup> Ginette Verstraete, “Media, Places, and Assemblies: From Videowalls to ‘Smart’ Environments,” *Space and Culture* 19, no. 4 (2016): 312.

<sup>46</sup> Lev Manovich, “The Poetics of Augmented Space,” *Visual Communication* 5, no. 2 (2006): 220.

<sup>47</sup> Ethan Tussey, “Connected Viewing on the Second Screen: The Limitations of the Living Room,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 204.

labor, while ignoring mobile possibilities in hopes of re-entrenching familiar ways of engaging with the televisual.<sup>48</sup> Stated differently, it's easy to forget these apps existed precisely because of how managed and top-down these trials in audience engagement truly were. Because many connected viewing apps required the use of television, there was still a focus on the audience and the living room instead of new ways that technology might work beyond the traditional context. Furthermore, while these apps all provide different ways of watching, they did mirror the traditional TV experience; even "extended content" for television programs was not always designed for mobile use.<sup>49</sup> Even for a platform like Netflix, in which corporate leadership appears keenly aware of growing mobile habits, advertisements for the service often depicted the living room as the site of streaming. What the rise (and fall) of second-screen applications for use alongside television programs establishes, then, is that the living room remained an important imagined site of televisual engagement—even as audiences increasingly began engaging with mobile devices to stream.

The iPhone, previously noted as a key milestone in the history and development of mobile streaming, is now well over fifteen years old. As mobile internet infrastructures improved from 3G to far more powerful 4G and 5G connectivity, individuals and platforms embraced streaming in a variety of spaces never before possible. By extension, a large part of the mobile streaming shift is interaction with new forms of the screen that may be untethered from the domestic spaces of the televisual. Key here is the importance of televisual screen size, noting that the notion of the television "set" has disappeared. We now speak of screens, monitors, and platforms, with a new focus on functionality and technical interdependence.<sup>50</sup> As emphasized in the discussion of a changing definition of flow, Jeremy Packer and Kathleen Oswald call for more attention to the history of screen technologies in relation to mobile media, noting how screen technologies function as a method of networked control exerted by and through each node, as mobile users move through space.<sup>51</sup> Mobile screens, then, are part of ongoing types of biopolitical management, as the role of the screen in public life becomes more ubiquitous and mobile.<sup>52</sup> As this brief subset of

<sup>48</sup> Tussey, "Connected Viewing," 214.

<sup>49</sup> Tussey, "Connected Viewing," 210.

<sup>50</sup> Paul Frosh, "The Face of Television," in *The Screen Media Reader: Culture, Theory, Practice*, ed. Stephen Monteiro (Bloomsbury, 2017), 307.

<sup>51</sup> Packer and Oswald, "From Windscreen to Widescreen," 336.

<sup>52</sup> Sean Cubitt, "Current Screens," in *The Screen Media Reader: Culture, Theory, Practice*, ed.

literature on screens shows, investigations of the screen often examine the materiality of the screen in relation to social and cultural forms, as well as questions of power and control.

Inspiring to our own work is recent scholarship from Barbara Maly-Bowe, who examines how Netflix influences the relationship between media consumption and the concept of the home. Similarly coming from a place of reinterpreting Raymond Williams's notion of mobile privatization, Maly-Bowe introduces the concept of "private mobilization." Here, Netflix redefines the home in three key ways: (1) It disrupts and reclaims traditional associations between television, mobility, and domestic life; (2) It links personalization and diversity to a sense of belonging, using "home" as a narrative tool for inclusion; and (3) It commodifies the private sphere, aligning with neoliberal market forces by transforming home into a space of both consumption and surveillance. Netflix, then, fosters a sense of personal belonging through diverse storytelling and flexible viewing, while also reinforcing corporate control over personal data and cultural narratives. We've examined elements of this increased corporate control in previous chapters. In addition, our own mobile personalization emphasizes the longstanding desire to monetize and make productive new times and spaces via mobile and portable technology.

Relevant overall in this section is that there is an enduring relationship between screens, space, and audio-visual entertainment. As explained in chapter 1, the screen of the cinema and its precursors structured urban and rural spaces. The televisual, by contrast, was a far less public but incredibly critical technology, introduced into the domestic space of the home at a time when families were becoming more mobile and withdrawn. The screen as technology has continued for decades to structure public spaces of stores, offices, schools, and more. However, the mobile screen of the smartphone allows a new orientation to space—one that is commodified and personalized. If Maly-Bowie focuses on private mobilization in terms of how the home is redefined by Netflix and other platforms, we instead want to trace the trajectory of mobile personalization. Key here is how the mobile screen of the smartphone can be used to transform all kinds of times and spaces for the user, while also working toward monetization and commodification via data capture. Relatedly, practices of mobile streaming animate scholarship on mobility and hybridity of space.

## Mobile and Hybrid Streaming

For decades, the dominant vision of the television operated within the confines of the home and the imagined (and real) site of the living room. The mobile aspect of the televisual enters in the form of imagined mobility. Audiences were able to “travel” around the world thanks to new audio-visual coverage of places, people, and events previously unseen and unheard. Of course, as we examined in chapter 1, there were simultaneously mobile and portable orientations to the televisual and the cinematic that coexisted with the prevailing vision of static and domestic television. Our overarching argument in that chapter is that there is a long lineage of a mobile and portable desire for entertainment and information in a variety of spaces and times via technology. Viewed from that lens, mobile streaming is a continuation of an existing trajectory and not something wholly new.

The “mobile” is a concept that’s at once separate from but also deeply intertwined with space. The prior section discussed the spatial turn, a moment of increased scholarly attention to spaces as multiplicitous, relational, and always under construction. In the 1990s, the mobilities turn represented another moment of scholarly emphasis, this time on the increasingly important dynamic of mobility. The 1990s were a period of greater global connection, driven, in part, by new international trade agreements, the fall of the Soviet Union, and the nascent commercial web. Scholars interested in the understudied element of mobility to local and global social formations emphasized questions of globalization, the production and reproduction of territories and nation-states, as well as the implications for identities and belonging.<sup>53</sup> Other key mobility theorists highlight how mobilities are intertwined with the social. Mobility simultaneously produces and is produced by social relations.<sup>54</sup> Furthermore, mobility studies include the act of physical movement, representations of movement that give mobility shared meaning, and the experienced and embodied practice of movement.<sup>55</sup> As such, “the mobile” is not only the physical act of movement, but also how movement is represented in cultural forms, as well as the ontological and phenomenological dimensions of mobility that require attention.

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<sup>53</sup> Mimi Sheller and John Urry, “The New Mobilities Paradigm,” *Environment and Planning A* 38, no. 2 (2006): 211.

<sup>54</sup> Tim Cresswell, “Towards a Politics of Mobility,” *Environment and Planning D: Society & Space* 28, no. 1 (2010): 21.

<sup>55</sup> Cresswell, “Towards a Politics of Mobility,” 19.

What is important about the mobilities turn is that a conception of space and the social require attention to movement itself, which has implications for technologies and practices that include mobile streaming. Stephen Wiley and Jeremy Packer clarify that the mobilities turn allows scholarship to move away from a media-centric approach (and the problematic questions of media effects, technological determinism, and audience agency). Instead, emphasis on questions of spatial production and mobilities allows scholars to situate communication technologies and discursive frameworks within a broader social field that encompasses networks of circulation, mobilities and immobilities, infrastructures surveillance control, and social organization.<sup>56</sup>

Taken together, the mobilities turn is critical to the study of streaming media for several reasons. First, the prevalence of mobile streaming might not “replace” fixed streaming forms, yet users around the world are increasingly engaging with the mobile, with younger individuals heavily gravitating toward portable and mobile streaming. Second, the new forms of streaming possible via mobile technologies and infrastructures—not only watching cinematic and televisual content but also captured and shared mobile streaming content—requires an expansion of what it means “to stream.” To this end, the spatial, mobile, and portable dimensions of streaming make mobile streaming a critical object of scholarly examination.

Examples show how location and mobility have become key aspects of new streaming practices. One study of early developing mobile engagement found that viewing habits of university students were strongly shaped by the location of viewing.<sup>57</sup> Other similarly early studies found that different platforms are amenable to different viewing styles, finding that mobile media is more participatory and “lean forward” in terms of viewing practices and content consumption.<sup>58</sup> This active or passive demarcation mirrors the conceptualizations of streaming as fixed versus mobile dialectic and is increasingly relevant to consider when taking into account streaming not only through audio-visual content platforms but also mobile streaming and livestreaming through social media. As mobile streaming takes on new forms, including streaming through social media, it is critical to emphasize

<sup>56</sup> Stephen B. Crofts Wiley and Jeremy Packer, “Rethinking Communication After the Mobilities Turn,” *The Communication Review* 13, no. 4 (2010): 264.

<sup>57</sup> Chuck Tryon and Max Dawson, “Streaming U,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 218.

<sup>58</sup> Sharon Strover and William Moner, “The Contours of On-Demand Viewing,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 248.

how space and mobile devices remediate a static history of streaming largely connected to the fixed and domestic spaces of the televisual. In other words, mobile streaming expands these activities to new times and spaces that were previously only imagined or perhaps just briefly experimented with.

Finally, the mobile nature of contemporary streaming practices further complicates divides between physical and digital spaces. Scholarship and popular culture depictions in the 1990s and early 2000s positioned the web as a “cyberspace” that individuals would enter or leave via fixed devices.<sup>59</sup> Accompanying this bifurcation between the digital and physical were prognostications that included the dissolution of cities, the reduced need for mobility, spatial homogenization, and the unimportance of the corporeal body.<sup>60</sup> Predictions about the future of technology often get it wrong, and these instances are no exception. Instead, mobile streaming is best considered as part of hybrid spaces, merging the physical and digital when it comes to social environments and created in large part by the mobility of individuals connected via mobile devices.<sup>61</sup> Hybrid spaces examine the mutual imbrication of the digital and physical via location-aware mobile technologies. This concept of hybridity is a crucial departure of mobile streaming from the televisual, cinema, and even more fixed forms of streaming. The aspects of mobility, portability, pervasive web connectivity, and location-awareness make mobile streaming a novel media practice with substantial implications for how the user senses, experiences, and moves through space. This is also a major question of power—specifically, how mobile streaming and web-connected screens guide the user through space and make productive their time.

This brief overview of space, mobility, and hybrid space foregrounds the importance of space as a relational, ongoing, and multiplicitous construction with social and communication dimensions. Moreover, mobility and movement—both physical and digital—are important aspects of the construction and reconstruction of spaces as hybrid. The spatial and mobile turns, as well as understanding the elision between the physical and digital via mobile technologies as a form of hybrid space is critical to understanding mobile

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<sup>59</sup> Helen Couclelis and Harvey J. Miller, “Misses, Near-Misses and Surprises in Forecasting the Informational City,” in *Societies and Cities in the Age of Instant Access*, ed. Harvey J. Miller (Springer Netherlands, 2007), 72.

<sup>60</sup> Francis Cairncross, *The Death of Distance: How the Communications Revolution Is Changing our Lives* (Harvard Business School Press, 1997), 10.

<sup>61</sup> Adriana de Souza e Silva, “From Cyber to Hybrid: Mobile Technologies as Interfaces of Hybrid Spaces,” *Space and Culture* 9, no. 3 (2006): 262.

streaming. Mobile streaming practices take a variety of forms, yet it is the new connections to and constructions of space, the movement through space in concert with audio-visual streaming content, and the hybridity of space due to data collection and the new relevance of location, geographic, and device data to streaming that require new attention.

### *Mobile Streaming, Big Data, and Algorithms*

It's a story told so many times that it's become streaming legend. Netflix's "gamble" on rebooting *House of Cards*, one of the platform's early hit series, was not a risk in the traditional sense. That is, the development cycle for Netflix's original programming heavily departed from the production cycle of television networks. When networks relied on some level of spotty ratings data and past trends, the decision to greenlight or pass on a show was largely driven by gut instinct. Netflix, conversely, had gobs of audience data points to work with. The company knew there was an audience who loved the original *House of Cards*, they knew there was an audience who would watch actor Kevin Spacey, and they knew there were dedicated fans of director David Fincher. What mattered was looking at points of overlap between these groups to create some level of empirical evidence that an audience existed for a new series.<sup>62</sup> In other words, big data has been key to content production in the streaming era.

Perhaps the most significant departure of streaming media from preceding forms of audio-visual engagement comes in the form of big data and algorithms. As previously mentioned in our focus on streaming industry shifts, streaming media is part of a mass media alteration from a transmission model to one that also includes forms of data collection, storage, and processing. This utilization of data is critical in understanding streaming media. Furthermore, the omnivorous nature of big data collection, as well as increasing engagement with mobile forms of streaming magnifies the importance of data points for device type, location, and mobility.

The ideology of collecting any and all data points for use in decision making and predictive analysis now undergirds just about any contemporary field—from streaming video to social media, from medicine to industrial production, and from government to corporate decision making. For streaming specifically, the switch from analog to digital cable, then to DVR, and

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<sup>62</sup> Blake Hallinan and Ted Striphas, "Recommended for You: The Netflix Prize and the Production of Algorithmic Culture," *New Media & Society* 18, no. 1 (2016): 128.

finally to streaming permitted a “quantum leap” in tracking and monitoring user behavior.<sup>63</sup> Early on, this included what shows audiences stored on their DVRs, when they viewed content, when they paused, rewound, or fast-forwarded. This data was supercharged by streaming platforms like Netflix, which tracks what we watch, when, where, with what device, when we pause, rewind, or skip ahead, and when we abandon a show entirely.<sup>64</sup> Consequently, Netflix and other streaming platforms can parlay user meta-data into decision making regarding original programming. Netflix, like other providers, allows user data to be shared between partners like internet service providers, mobile phone carries, and voice assistant platforms.<sup>65</sup> Because these corporations have access to viewer preferences regarding genre, actors, directors, binging penchants, and more, streaming providers can use algorithms to create programming with a built-in, near-guaranteed audience. Algorithms, moreover, deliver and promote content based on previous user activity, ensuring a more efficient, curated, and personalized relationship to audio-visual media, as well as the potential of a filter-bubble effect in terms of suggested content.

Big data and algorithms matter from a mobile streaming perspective as well. TikTok’s own recommendation algorithm is based on factors that include user interactions (videos that are liked or shared, followed accounts, comments posted), video information (captions, sounds, hashtags), and device and account settings (language preference, device type, country settings). Further, TikTok’s machine learning models primarily rely on metadata like description, tags, time, and location of video upload.<sup>66</sup> In essence, contemporary streaming practices build off the data structures of more fixed streaming antecedents; however, elements of location and device type become more magnified as streaming becomes mobile and social.

While “the algorithm” has become a bit of a catch-all buzzword by virtue of social and streaming media platforms, what the term refers to is merely a mathematical or logical set of instructions.<sup>67</sup> Layers of algorithms can be used to make sense of large datasets via what’s known as machine learn-

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<sup>63</sup> Mark Andrejevic, “Surveillance in the Digital Enclosure,” *The Communication Review* 10, no. 4 (2007): 308.

<sup>64</sup> Kate O’Flaherty, “All the Ways Netflix Tracks You and What You Watch,” *Wired*, October 10, 2021.

<sup>65</sup> O’Flaherty, “All the Ways Netflix Tracks You.”

<sup>66</sup> Will Knight, “This Technique Can Make It Easier for AI to Understand Videos,” *Wired*, October 15, 2019.

<sup>67</sup> Lucas Introna and David Wood, “Picturing Algorithmic Surveillance: The Politics of Facial Recognition Systems,” *Surveillance & Society* 2, no. 2–3 (2004): 180.

ing. And the use of neural networks constitutes a more powerful form of machine learning known as deep learning. Machine and deep learning are subsets of what we'd call artificial intelligence, most commonly used now to identify relationships, detect and uncover patterns, classify and transcribe, and even to decide or recommend in an automated fashion.<sup>68</sup> In turn, recommendation algorithms are critical to what we see when we stream—no matter if that occurs while watching Amazon Prime on a smart TV in one's living room or streaming TikTok on a smartphone while riding the metro.

Equally important is the aforementioned aspect of big data. Big data has been defined numerous ways via some key characteristics. Certainly, the idea of collecting and storing data about the world is nothing new. Even ancient Sumerians, the first human civilization, utilized record keeping. What sets the big data era apart includes the digitization of records for computational readability, unprecedented storage capacity to house data digitally, and new modes of algorithmic analysis to help make sense of data.<sup>69</sup> Big data allows entities to harness information in novel ways, producing insights of significant value. This entails the ability to analyze enormous amounts of data instead of smaller (likely randomized) sets; a willingness to embrace data messiness instead of precision; and a growing respect for correlations.<sup>70</sup> Finally, big data is omnivorous. Platforms attempt to collect any type of information they can from user practices and interactions for analysis and insights.<sup>71</sup> This omnivorous appetite of big data for ever-increasing types of data is increasingly relevant to mobile streaming, as device type, location, and geographic data add to this insatiable mode of data collection. Taken together, big data is as much an ontological shift as it is an epistemic one: This process entails an intensified digitization of the longstanding process of collecting, storing, and processing data, along with a reverence for a worldview that if we collect enough data, we can access and predict all aspects of reality itself. Big data is correlative and predictive, yet these correlations are often opaque to a point beyond human comprehension.

While big data and algorithms largely structure streaming media practices, it is worth noting the asymmetrical aspects of this platform-user relation-

<sup>68</sup> Kevin P. Murphy, *Machine Learning: A Probabilistic Perspective* (MIT Press, 2012); Ian Goodfellow et al., *Deep Learning* (MIT Press, 2017).

<sup>69</sup> Mark Andrejevic and Kelly Gates, "Big Data Surveillance: Introduction," *Surveillance & Society* 12, no. 2 (2014): 186.

<sup>70</sup> Viktor Mayer-Schönberger and Kenneth Cukier, *Big Data: A Revolution That Will Transform How We Live, Work, and Think* (Houghton Mifflin Harcourt, 2013), 12.

<sup>71</sup> Mark Andrejevic, "The Big Data Divide," *International Journal of Communication* 8 (2014): 1674.

ship. There exists a growing divide between those with access to big data and its attendant infrastructure and those without.<sup>72</sup> We hear about big data living in “the cloud.” The cloud merely refers to tons of servers housed in a warehouse owned by large companies. The terminology of the cloud invokes a lightness and freedom that runs contra to the reality of data collection; this metaphor obfuscates the surveillant aspects present in all forms of streaming media.<sup>73</sup> Put another way, platforms like TikTok have significant power, and this has resulted in geopolitical controversy when it comes to attempts at regulation. This situation remains ongoing and is covered thoroughly in other publications. Nevertheless, global bans of TikTok by a number of countries demonstrate how platforms vie for power via reach, users, data, and influence.

As scholars have noted, the conjunction between big data and algorithms has had significant impacts on what we watch, how we watch, and what gets made in the first place. We’ve mentioned in this section how big data collection is a far more efficient model of measuring viewership than previous forms when compared to prior systems of assessing television ratings through a variety of technologies used for representing, and calculating audiences, including face-to-face interviews, phone surveys, and diaries.<sup>74</sup> The use of big data has led to a shift from intuition-based decision making by studio executives to data-driven models in streaming platforms, where audience engagement continuously strengthens algorithmic predictions. These algorithms, though largely proprietary, are central to how platforms mediate cultural authority.<sup>75</sup> Blake Hallihan and Ted Striphas examine this dynamic through the case of *The Netflix Prize*, a rare look into how the Netflix algorithm operated through an open competition held by the company.<sup>76</sup> The authors explain that streaming algorithms often work to mediate more divisive outlier content; rather than evaluating cultural worth, Netflix moderates atypical works to fit broader patterns, displacing cultural judgment into technical processes. This reinforces a closed loop of recommendation, where content increasingly reflects user preferences rather than challeng-

<sup>72</sup> Andrejevic, “Big Data Divide,” 1674.

<sup>73</sup> Mark Andrejevic, “Surveillance in the Digital Enclosure,” *The Communication Review* 10, no. 4 (2007): 304.

<sup>74</sup> Jérôme Bourdon and Cécile Méadel, “Inside Television Audience Measurement: Deconstructing the Ratings Machine,” *Media, Culture & Society* 33, no. 5 (2011): 792.

<sup>75</sup> Michael D. Smith and Rahul Telang, *Streaming, Sharing, Stealing: Big Data and the Future of Entertainment* (MIT Press, 2016), 211.

<sup>76</sup> Blake Hallinan and Ted Striphas, “Recommended for You: The Netflix Prize and the Production of Algorithmic Culture,” *New Media & Society* 18, no. 1 (2016).

ing them. Consequently, cultural value is no longer anchored in human interpretation alone; systems now play a decisive role in what enters and shapes the cultural field.

Of course, this book is all about the mobile aspects of streaming media. Both big data and algorithms take on new forms via mobile practices. As previously noted, mobile devices initially became central to streaming media through the rise of the second-screen experience. The second screen facilitated increased real-time monitoring, customization, and targeting—key elements of the interactive commercial economy.<sup>77</sup> Framed as enhancing viewer agency, second-screen apps support a “push” model of data collection, capturing user input through designed interactions. This model reaffirms the value of live viewing and advertising while contributing to a broader system of data capture.<sup>78</sup> Notably, this viewing paradigm was grounded in the domestic space of the living room, where mobile devices are woven into fixed media habits. Other studies have found that college student viewing habits are shaped by place, social setting, cost, and access, signifying that *what* and *how* students watch are factors of *where* they watch.<sup>79</sup>

Similarly, other scholars underscore the role of mobility, temporality, and platform in shaping viewing behavior. They note that mobile viewing tends to be more participatory and “lean forward,” necessitating a rethinking of viewership in terms of devices, platforms, and context.<sup>80</sup> Together, these accounts highlight the entanglement of mobile streaming with big data and algorithmic governance. Streaming platforms rely on pervasive data collection to optimize both content production and targeted distribution. As audiences increasingly stream via mobile devices, data tied to geography, behavior, and device becomes critical to predictive analytics, reinforcing the commercial imperative for constant growth. In essence, mobile streaming represents an emergent frontier of data capture, particularly because of how well it captures user attention across new times and spaces.

Taken together, big data and algorithms are pervasive aspects of both streaming and mobile steaming media. Big data and algorithms have been a significant factor in the more efficient production of content as well as content distribution via suggestion to targeted audiences by streaming

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<sup>77</sup> Lee and Andrejevic, “Second-Screen Theory,” 42.

<sup>78</sup> Lee and Andrejevic, “Second-Screen Theory,” 45.

<sup>79</sup> Tryon and Dawson, “Streaming U,” 248.

<sup>80</sup> Sharon Stover and William Moner, “The Contours of On-Demand Viewing,” in *Connected Viewing: Selling, Streaming and Sharing Media in the Digital Age*, ed. Jennifer Holt and Kevin Sanson (Routledge, 2014), 248.

platforms. Moreover, the goal of continual profit expansion aligns with the omnivorous mode of big data collection: More data allows for more correlative predictive potential, and device, geographic, and location data are increasingly important as individuals engage in mobile streaming practices.

## *Conceptualizing Mobile Streaming*

In this chapter, we've engaged with a major question: How does mobile streaming remediate prior understandings of flow, liveness and sharedness, screens and space, mobile and hybrid, and algorithms and big data? This is far from a one-size-fits-all situation, especially considering just how embedded streaming has become. Immensely popular mobile-focused platforms like TikTok and Instagram operate alongside audio-visual media providers such as Netflix, Amazon, and Hulu. With users more and more likely to engage in streaming practices via a mobile device, it's critical to consider how the increase in user captured and shared streaming content, designed for the short-form temporal duration and comparatively small screen size of mobile viewing, necessitates reexamination of what it means to stream.

As we've highlighted in previous chapters, there's a historical dimension at play. The desire for a mobile and portable orientation to entertainment and information anytime and anywhere is far from a new goal. While historical examples help situate this trajectory, we've also discussed changes in the material technologies and infrastructures needed to enable the kinds of mobile streaming engagement we've come to integrate into our lives. And finally, we've looked at the platform model as key to changing mobile streaming monetization.

This trajectory has led us to this final chapter in which we've reconsidered how scholarly accounts on televisual, cinematic, and fixed streaming are challenged, remade, and remediated by mobile and portable streaming. These concepts, deeply intertwined with pre-streaming mass media and fixed streaming alike, are found just as readily in mobile streaming practices. Nevertheless, the myriad of novel ways to stream anytime and anywhere does necessitate a critical reassessment of the unbroken experience of watching, of how mobile streaming is live and shared like never before, how mobile screens are part of the spaces where we watch, how streaming involves movement and hybrid orientations to space, and how new and old forms of data and algorithms are part of these experiences.

In addition, the temporal, flexible, and rhythmic aspects of streaming and mobile streaming make clear that the concepts of liveness and sharedness that largely structured televisual and other synchronous mass media engagement continue to be part of streaming media as well. While some of these features have been abandoned by the audio-visual streaming providers commonly associated with streaming (Netflix, Amazon, etc.), the adoption of and increased emphasis on livestreaming by social media platforms demonstrate the attention paid to a shared streaming experience. While the importance of big data and algorithmic engagement has long structured the production and economic efficiency of streaming, the capture of location, geography, and device information is critical to new evolutions of mobile streaming. What is critical here is that location-aware features are gradually made salient within new mobile streaming alternatives.

Together, this chapter has contextualized new forms of streaming media by taking into account novel mobile streaming practices. Although more fixed forms of streaming media have not been rendered irrelevant nor replaced by mobile streaming, the growing amount of mobile engagement—including the overtaking of fixed streaming by mobile—necessitates increased scholarly engagement with this streaming evolution. Consequently, this chapter examined the increasingly mobile dimensions of streaming regarding some affiliated sociotechnical practices such as industry shifts, liveness, sharedness, plasticity, and algorithmic engagement. These are all elements of mobile personalization. Mobile and portable technologies are critical components in the ongoing transformation of times and spaces, as they provide entertainment and information for users while simultaneously operating toward the goals of monetization and control. Critical here is that the local is not erased but made salient through mobile devices, infrastructures, streaming practices, location-aware technologies, and data collection. Taking seriously the shift to mobile streaming means reconsidering the increasingly dominant method of audio-visual media interaction as a dynamic rather than static process in which location and geography matter.